04/25

Life Processes

Those activities, which help us to continue life, are life processes.

→ Certain activities which are common to both living and non-living, are not considered as life processes.

→ Certain activities which are performed by living beings, but is not required for living, are not included under life processes.

WHAT ARE LIFE PROCESSES?

- Intake of nutrition
- Respiration
- Removal of waste

Life processes are necessary for living beings irrespective of body size, design & number of cells present.

* DIFFUSION- Movement of molecules of any substance (whether solid, liquid or gas) from a level of higher concentration to a level of lower concentration with or without a semi-permeable membrane.

* OSMOSIS- Movement of water molecules only from a level of higher concentration to a level of lower concentration through a semi-permiable membrane, until & unless an ~~equiberalium~~ equilibrium is achieved.

NUTRITION

Nutrition is intake of food in proper amount which satisfies the nutritional need of a body. It includes- Carbohydrates, Vitamins, Minerals, fats, Roughage / Fibre, Protein & Water.

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→ Allmentary canal shows a sythmical movement called peristaltic movement.

→ This process is called persistalsis.

→ This helps the food to pass through the alimentary canal, downward.

→ Stomach is an elastic bag like structure.

→ It expands, as the food enters.

→ The stomach has digestive juices, which helps in digestion of food.

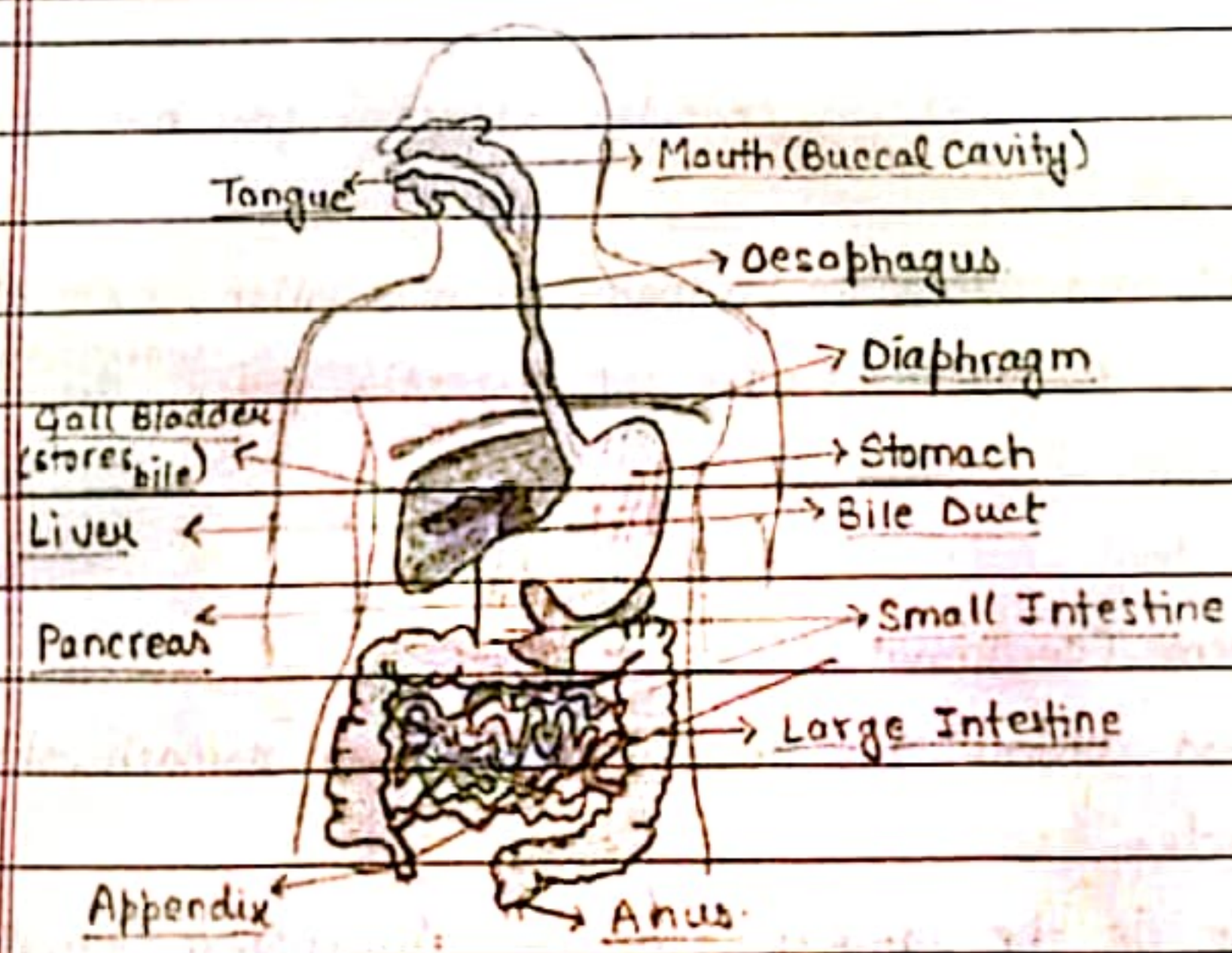
→ Wall of stomach produces digestive enzyme.

Stomach produces

HCl	Pepsin	Mucus
– Helps pepsin to work.	– Helps in digestion of proteins.	– Protects the wall of Stomach from HCl (PH 1.4)

→ Pepsin can work only in acidic medium.

→ In case there is lack of food in stomach, body feels acidity & also develops stomach ulcers.



HUMAN ALIMENTARY CANAL

01/05/25

→ NUTRITION IN HUMAN BEINGS

Mouth
↓
Buccal Cavity
↓
[Voice Box] Larynx
↓
[Opening to Food Pipe] Pharynx — Epiglotis
↓
Oesophagus
↓
Stomach ← Liver, Gall Bladder, Pancreas
↓
Small Intestine
↓
Large Intestine
↓
Rectum
↓
Anus [Removal of waste]

→ The digestion of food starts from Buccal Cavity.
→ The food we intake are of different types.
→ Before we engulf food, all forms are mixed with saliva and crushed with teeth & all the food is then brought at similar form.
→ Saliva is produced by salivary gland.
→ Saliva contains enzyme salivary amylase.
→ Tongue is the most strongest muscle of the body.
→ Tongue rotates the food in the buccal cavity, so that food is mixed with saliva.
→ Complex sugar $\xrightarrow[\text{amylase}]{\text{salivary}}$ simple sugar.

- Plants require water & minerals for living.
- Both they get from soil
- Nutrients
  - Nitrates → present in soil, is dissolved
  - Nitrites → in water, before taken up.
  - Ammonia → by plants.

- Heterotrophic Nutrition
  - different
  - food.

- Those, who can not produce their own food.
- They depend on others for food, Show heterotrophic mode of nutrition.

→ Holozoic Nutrition

Intake / injestion of food, digestion of food, absorption of nutrien from food by the body & removal of the waste.

Lecture quee

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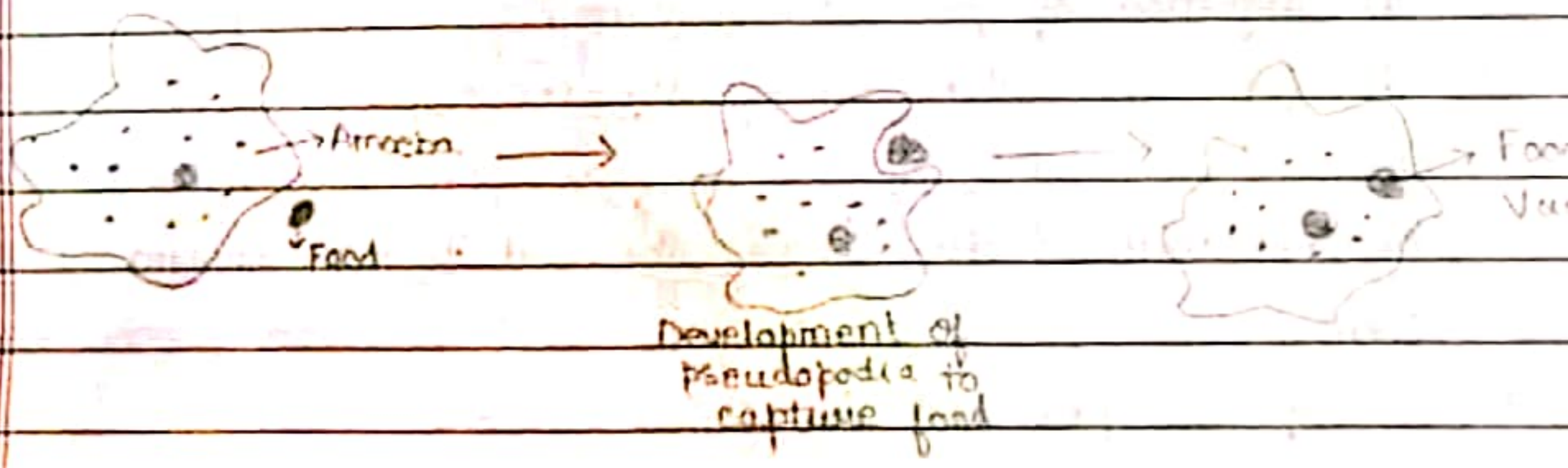
→ Amoeboid Nutrition



ENZYMES: Those chemical substances which do not participate in any chemical reaction but allow the rate of reaction are known as catalyst. Such catalyst when found inside any living system are known as Bio-catalyst. They are termed as Enzymes.

$A + B \xrightarrow{10\,min} C + D$

$A + B \xrightarrow{5\,min} C + D$

$A + B \xrightarrow{15\,min} C + D$ } catalyst → Biocatalyst → Enzymes (all are the same)

- Autotrophic Nutrition — auto - self, trophic - nutrition.

Organisms which can produce there own food by the help of inorganic substances & chlorophyll present within them. The end product produced is complex organic Carbohydrate which produces energy.

REACTION OF PHOTOSYNTHESIS:

$$6CO_2 + 12H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

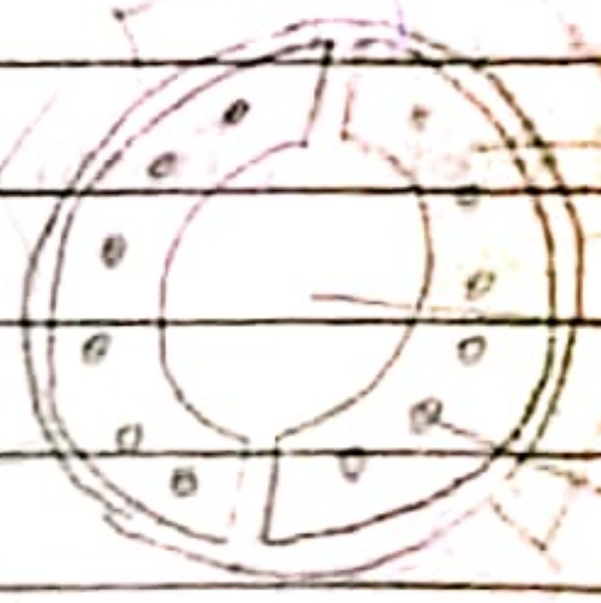
STEPS OF PHOTOSYNTHESIS:

(1.) Absorption of light energy by chlorophyll of leaves.
(2.) Conversion of light energy into chemical energy.
(3.) Splitting of H_2O molecule into hydrogen & oxygen.
(4.) Reduction of CO_2 into carbohydrate.

EXCHANGE OF GASES:

Stomata helps in exchange of gases & transpiration.

- In XEROPHYTIC Condition (desert condition), stomata are sunken (deep & closed), so as to avoid excess loss of water during day time.
- They have thick cuticle & waxy coating.



Closed Stomata/ Sunken Stomata

→ Intext Questions (NCERT Pg no - 81)

(Q1) What is criteria do we use to decide whether something is alive?

(Ans) The criteria we use to decide if something is alive is based on the presence of key life processes which generally include growth, metabolism, responsiveness to the environment, nutrition, movement etc.

(Q2) Why is diffusion insufficient to meet the oxygen requirements of multi-cellular organisms like humans?

(Ans) Diffusion is too slow to effectively transport oxygen over the long distances within large, multicellular organisms like humans. Our bodies are complex, and simple diffusion wouldn't be fast enough to get oxygen to all our cells.

(Q3) What are outside raw materials used for by an organism?

(Ans) Organisms utilize outside raw materials such as food (carbohydrate, fats, proteins), water, oxygen, minerals, carbondioxide and sunlight for energy, growth, metabolic activities etc.

(Q4) What ~~pr~~ processes would you consider essential for maintaining life?

Ans) Processes like: nutrition, respiration, transportation, excretion, metabolism etc. are essential for maintaining life.

Digestive System (continuation)

→ The digested food enters to small intestine from stomach, through sphincter muscle.

→ Small intestine is the largest part of alimentary canal.

→ The size of small intestine varies depending on the food.

→ The Herbivores have to digest, cellulose, which is difficult to digest, hence their small intestine is longer than carnivores.

Human Respiration

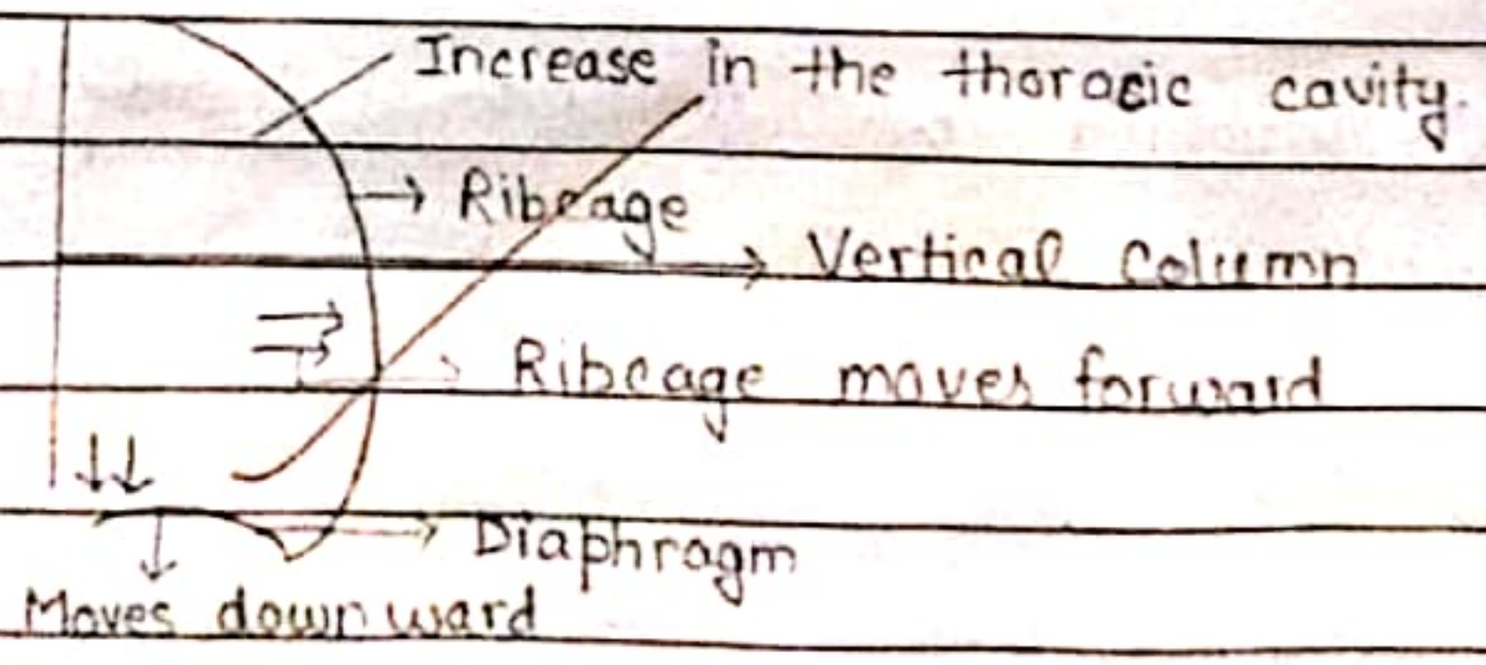
Nostril < nasal hair, mucus] traps dust during inhale of air
↓
Nasal chamber
↓
Larynx — voice box
↓
Bronchus (common passage)
↳ 1° Bronchii → 2° Bronchii → 3° Bronchii
↓
Bronchiole
↓
Alveoli (elastic sac like structure) → exchange of gases.

} LUNGS

→ The amount of air present in atmosphere is much more than the amount of air present in water.
→ The Aquatic animals utilize this oxygen for breathing. As a result, the rate of breathing is high in aquatic animals as compared to terrestial animals.

Breathing:

* Nostrils have nasal hair and mucus to stop dirt.
→ They refrain ~~in~~ the entry of dust into resp. tract.
→ Trachea have rings of cartilage which protects it from collapsing.



DURING INHALE

Q5) How is the small intestine designed to digest food?
Ans) Has extensive surface area, length, folds, villi, microvilli, enzyme actions in order to have proper digestion.

→ RESPIRATION

- Breathing

→ Intake of air and removal of CO_2 rich air.

Respiration: The process of burning of food (digested & absorbed) to release energy in the form of ATP. This energy is utilised by the body at each & every cell. Hence, such a respiration is called Cellular Respiration.

O_2 + Digested + Absorbed Food $\xrightarrow{\text{Respiration}}$ Energy in form of ATP.
~~Glucose~~

Glucose in cytoplasm → Pyruvate + Energy
- → Aerobic @ mitochondria — CO_2 + H_2O + energy
- → lack of O_2 @ muscle cell — lactic acid + energy
- → anaerobic @ yeast — Ethanol + CO_2 + Energy

Aerobic Respiration = In sufficient supply of O_2.
Lack of Oxygen = In insufficient supply of O_2.
Anaerobic Respiration = Without O_2.

Amt. of O_2 involved $\propto$ Amount of Energy Produced.

→ In-text Questions (NCERT Pg-87)

(Q1.) What are the differences b/w Autotrophic and Heterotrophic Nutrition?

(Ans.)

FEATURE	AUTOTROPHIC	HETEROTROPHIC
Food Source	Produce their own food from inorganic substance	Obtain food by consuming other organisms or organic matter.
Energy Source	Light Energy or Chemical Energy	Chemical Energy derived from ingested organic compo
Nutritional Dependence	Self-Sufficient	Dependent on others.
Examples	Green Plants, Algae.	Fungi, Animals.

(Q2.) Where do plants get each of the raw materials required for photosynthesis?

(Ans.) Plants obtain CO_2 from atmosphere through leaves, water is absorbed from the soil, Sunlight is captured by chlorophyll within the leaves.

(Q3.) What is the role of acid in the stomach?

(Ans.)
- Aids in protein digestion.
- Activates pepsin.
- Kills harmful bacteria
- Helps in absorption of some nutrients.

(Q4.) What is the function of digestive enzymes?

(Ans.) To break down large, complex food molecules (carbohydrat proteins, fats) into smaller, simpler molecules that can be easily absorbed by the body into the blood stream.

Small Intestine digest:

- Carbohydrate
- Protein
- Fats

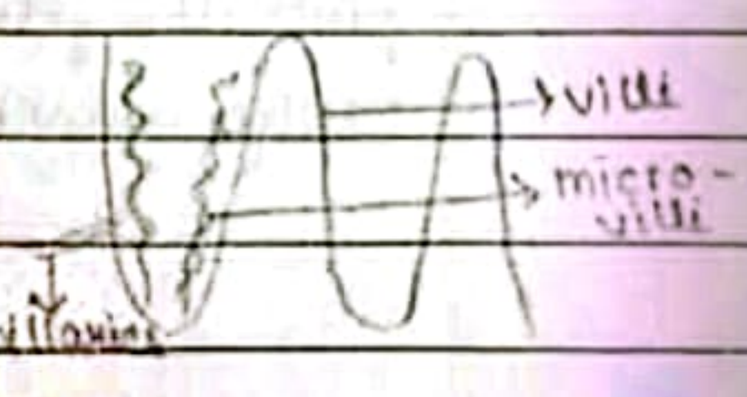
→ Pancreas, ~~Pancreatic juices~~ release pancreatic juices into small intestine.
→ Pancreatic juices can not work under acidic medium.
→ Bile juice (from Liver), convert the food from acidic to alkaline medium.
→ Bile juice also helps in digestion of fat.
→ It is difficult to digestion fats as they are of large globules.
→ These are first broken down to smaller fat globules.
→ Now, the bile salts digest small fat globules further.

Pancreatic Juices contains enzyme:
- Trypsin → digest protein
- Lipase → digest emulsified fat.

→ Final digestion in small intestine.

- Carbohydrate (complex) → Breaks down to simple sugar. (glucose)
- Protein → Breaks down to Amino acid
- Fat → Breaks down to
  - Fatty acid
  - Glycerol

→ Walls of small intestine have villi & micro villi.
→ They help in absorption of digested food.
The finger like projections is villi, which helps to increase the surface area, in a very small space.
→ Now, through small & minute capillaries, the absorbed food is transported to each & every cell of the body for various activities.
→ The absorbed food is transported to large intestine.



→ Large intestine absorbs only water.
→ Rest is moved to Rectum.
→ Through anus, the waste is removed, by anal sphincter.

→ Oxygenated blood moves from lungs to left auricle.
Auricle.

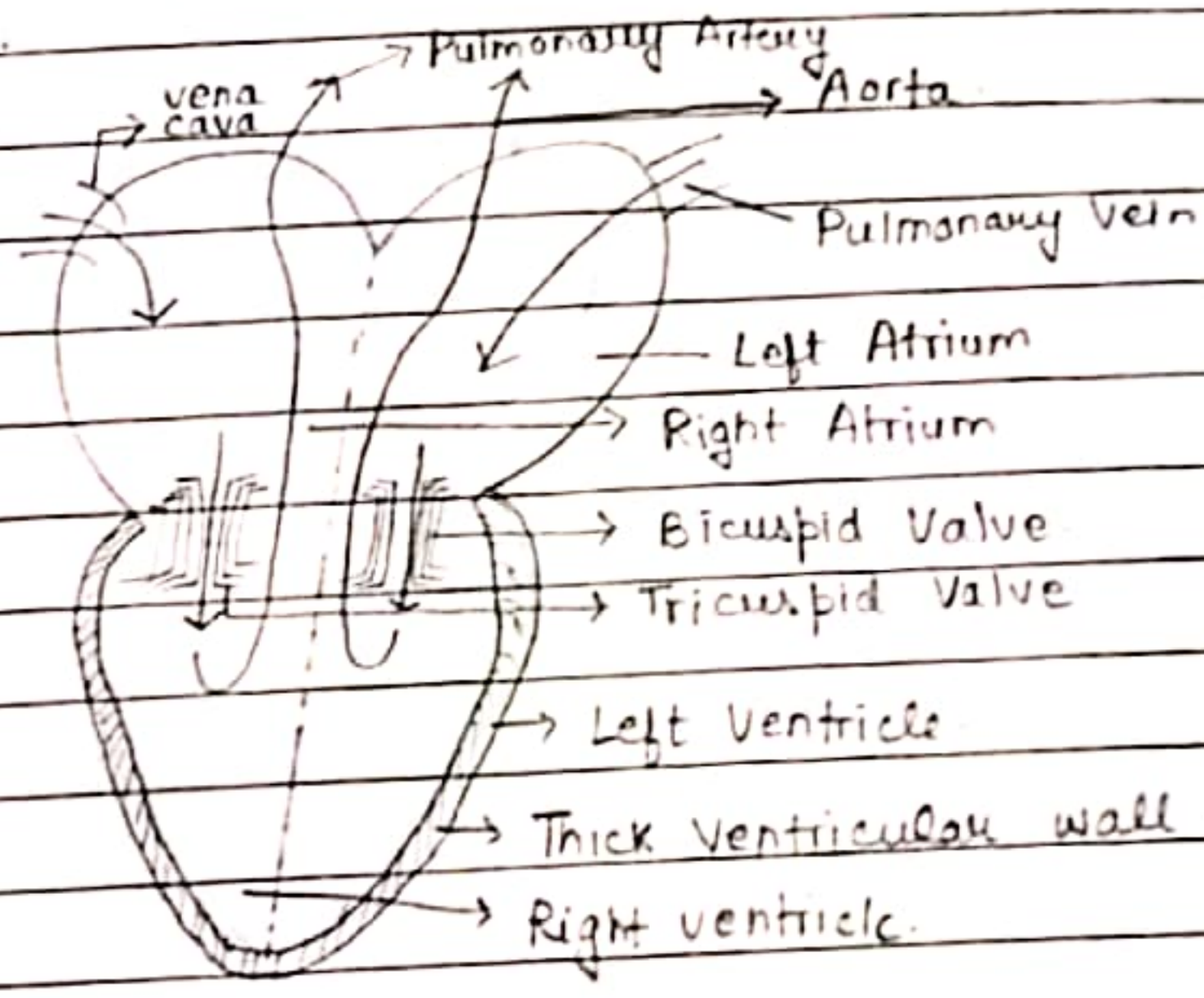
→ Oxygenated blood moves from left ~~au~~ auricle to left ventricle.

→ Oxygenated bloods moves ~~from~~ to body with the help of Aorta

→ From body, deoxygenated blood is carried by vena cava to Right Auricle.

→ From Right Auricle, the blood moves to Right Ventricle.

→ With the help of pulmonary artery, deoxygenated blood moves to lungs.



- VALVES

→ They make sure that there is no back flow of the blood within the heart or in the body.

→ The valves of ventricle is thicker than auricles because blood comes with a huge pressure from auricles.

- AURICLES

- ARTERIES

→ Arteries are tubes which carries oxygenated blood from heart to different parts of body.

→ Oxygenated blood moves from lungs to left auricle.
~~Auricle~~

→ Oxygenated blood moves from left ~~au~~ auricle to left ventricle.

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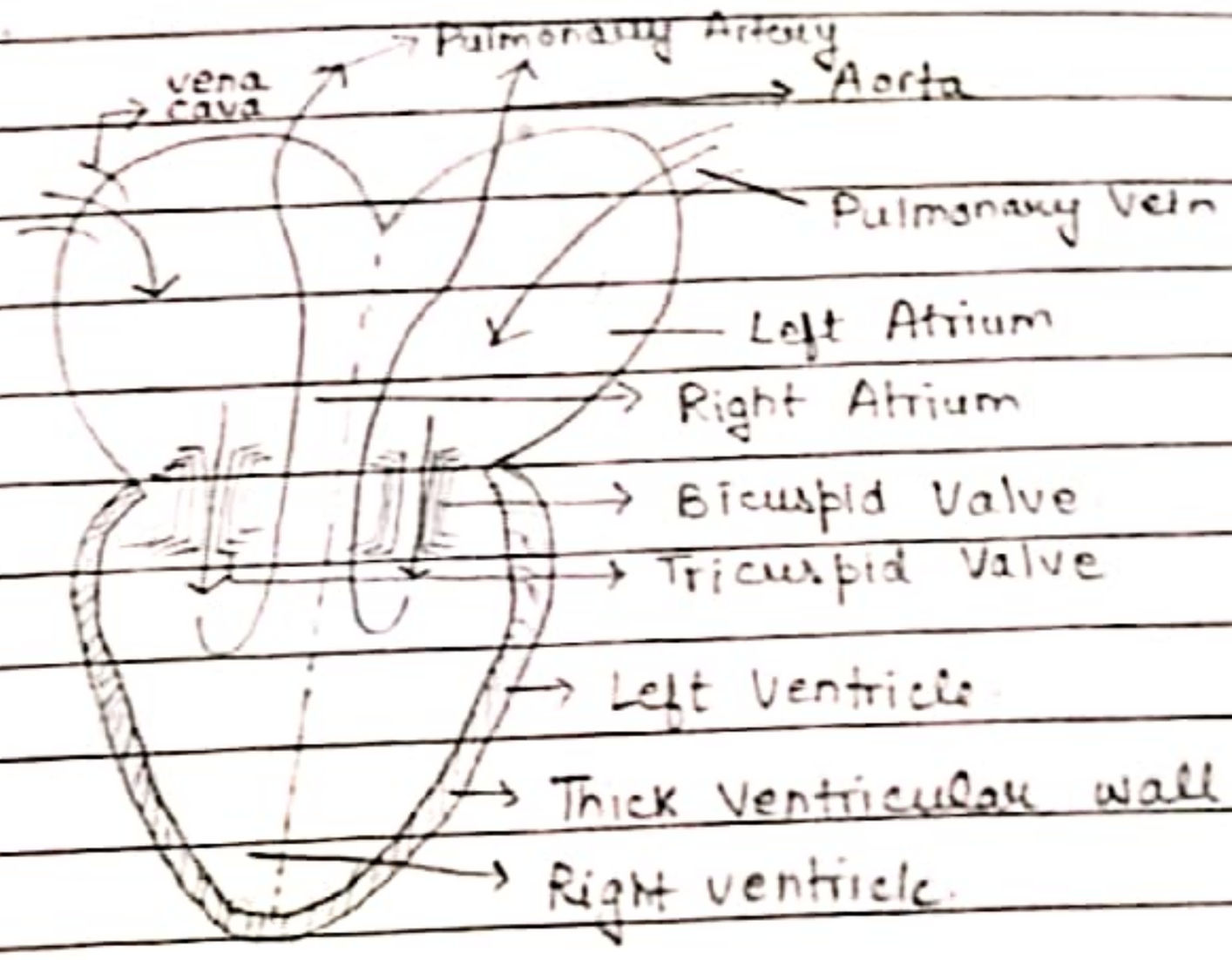

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- ~~AURICLES~~
- ARTERIES
→ Arteries are tubes which carries oxygenated blood from heart to different parts of body.

- Transportation.

TRANSPORTATION IN HUMAN BODY

→ Blood is a matrix fluid of plasma.

→ Blood transports material from one place to another within body.

→ Blood helps in transport of:

- Food
- Gases
- Waste Product.

→ This transport system also helps in repair of damaged portions within the body.

- OUR HEART: → It is the mesodermally originated, muscular organ

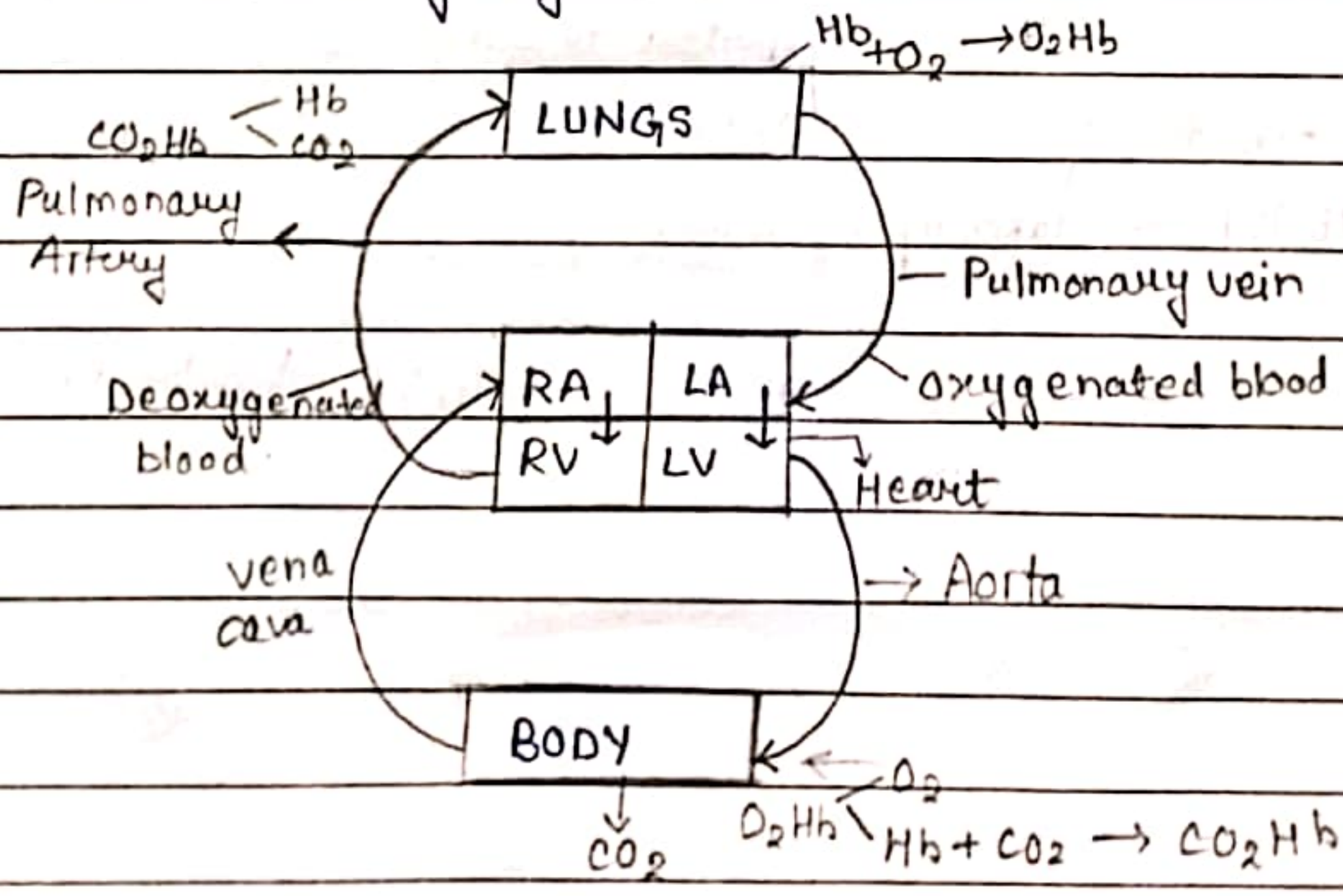
(meso → middle; derm → layer / skin)

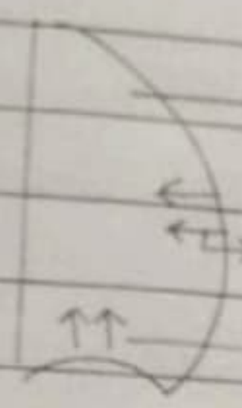
→ It is placed slightly left side in b/w two lungs in the thoracic cavity. (chest region)

→ It is as big as our closed left hand fist.

→ It has four chambers to avoid mixing of oxygenerated and deoxygenated blood.

- Double Circulatory System





Decrease in the thoracic cavity

ribcage moves backward

Diaphragm moves upward

DURING EXHALE

Both during inhale & exhale, there is some residual O_2 and CO_2 remaining in the lungs.

→ Respiratory pigment in human body is Haemoglobin [Hb]

→ It is found in the RBCs of blood

Haemoglobin
- Iron
- Protein

$Hb + O_2 \rightleftharpoons O_2Hb$ (oxyHb)

@ alveoli

↓ Transported to Body

$Hb + O_2 \rightleftharpoons O_2Hb$

- Hb is free
- O_2 taken up by body

$Hb + CO_2 \rightleftharpoons CO_2Hb$ (carbaminoHb)

@ body

↓ Transported to Lungs

$Hb + CO_2 \rightleftharpoons CO_2Hb$

- Hb is free
- CO_2 Taken up by alveoli

* $Hb + CO \longrightarrow COHb$ (carboxyHb)

- CO: Toxic

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- VEINS

→ Veins carry deoxygenated blood from different parts of the body to heart.

- CAPILLARIES

→ Capillaries are fine tubes which carry blood to the finest part of our body.

- PLATELETS

→ They are present in blood.

→ Help in blood clotting, incase of injury.

→ In an open wound, these platelets come in contact with atmospheric oxygen, which leads to formation of a thin layer, leading to stoping of blood flow.

Blood → RBC
↓
Hb

Rust ← [humidity + Fe] — Hb — globin (protein content)

- LYMPH

→ These are colourless component of blood called WBC.

→ They contain leucocyte cells.

→ They contain enzyme prothrombin and thrombin, which helps in fighting with the diseases. (in disease in body)

→ They form the immunity of the body.

→ They together contribute to lymphatic system of the body.

- Selective Reabsorbtion.

Intial filtrate – glucose, amino acids, salt, major amount of water are selectively reabsorbed

- Tubular secretion

→ Urine formed in Kidney is connected/converted to a long tube like structure Ureter.

→ Ureter is connected to urinary bladder.

→ Urine is passed out as there is an urge to urinate.

→ The pressure is exerted by muscular bladder.

→ Urethra passes the urine out.

→ Urinary bladder is under the control of Nervous System as it is a muscle.

- STRUCTURE OF NEPHRON

→ Nephron consists of:

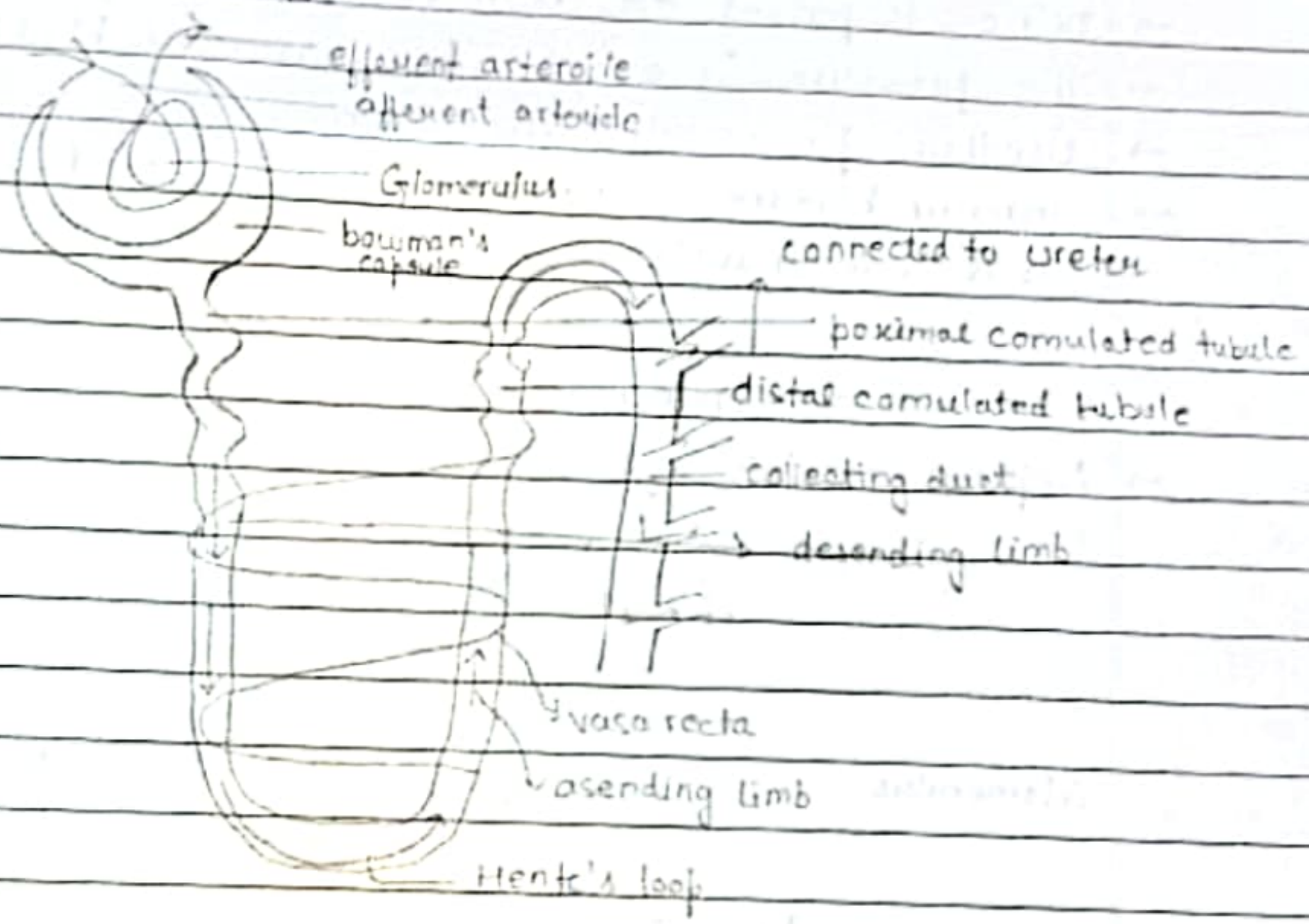
Components of Nephron

- Glomerulus
- Browman's capsule
- Malphigian Body.
- ascending limb
- descending limb
- proximal convulated tubule
- distal convulated tubule
- Henle's loop
- collecting duct.

FUNCTION OF KIDNEY:

→ Filters blood + osmoregulation (removal of toxic substance)
→ Removes N_2 waste from blood. Eg.: Urea and uric acid.
→ Functional unit of Kidney is Nephron.
→ Kidney performs ultrafiltration.
→ It selectively reabsorbs.
→ It shows tubular secretion.

- ULTRA FILTERATION



The cup shaped bowman's capsule and the capillaries (thin walled blood capillaries) glomerulus together forms Malphegian tubule which forms the filtrate which is collected in the collecting duct. And the process is called ultrafiltration.

Glomerulus + Browman's capsule = Malphegian Tubule.

Renal organ = Kidney
Artery - N_2 waste blood = Renal artery (unfiltered blood)
Vein - Blood after filter = Renal vein (filtered blood)
Amount of reabsorbtion depends on amount of excess blood available to the body.

→ In phloem, transport of food i.e. translocation, required energy in the form of ATP

EXCRETORY SYSTEM

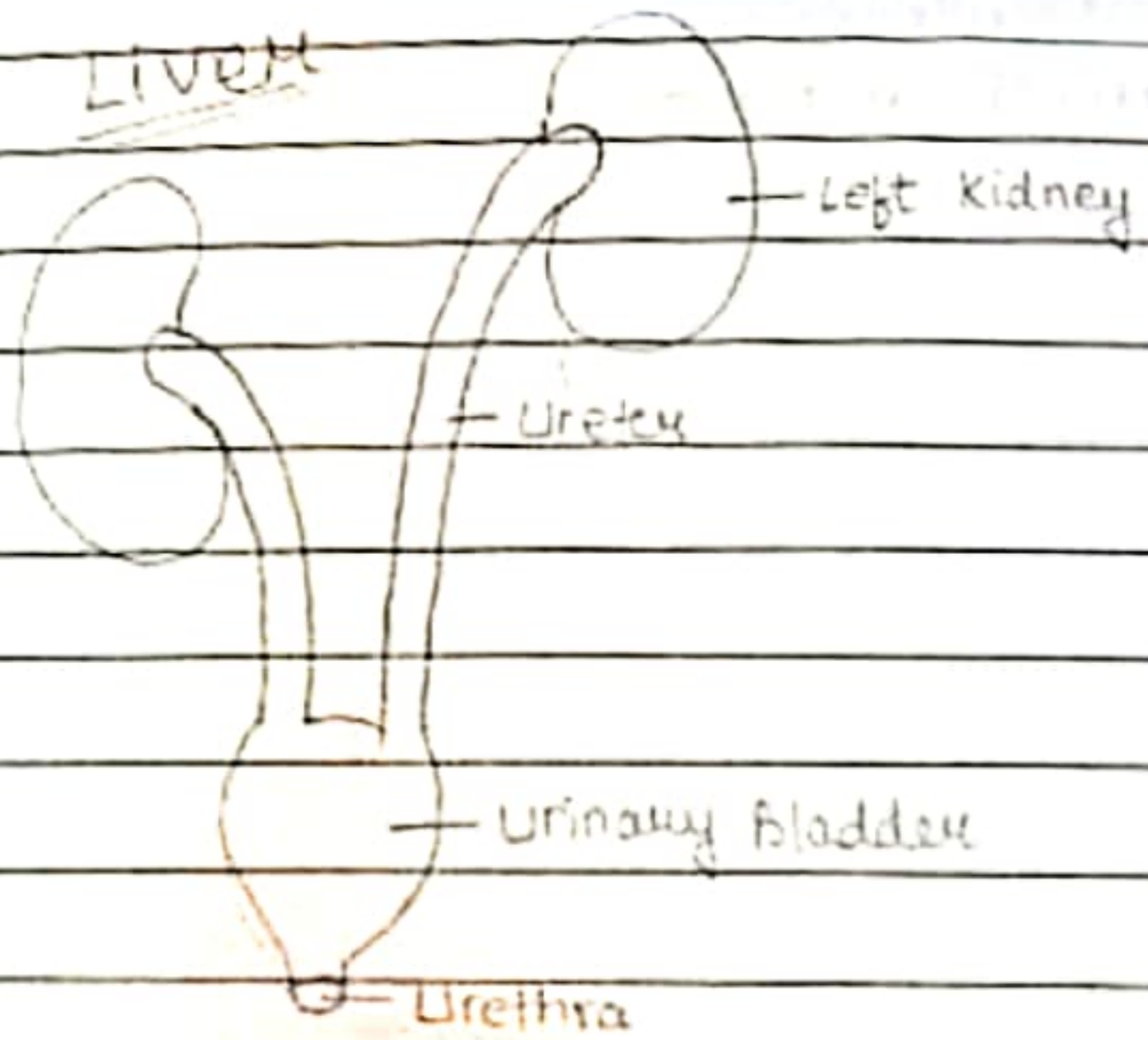
→ The process which helps in removal of waste in our body is taken care by excretory system.

→ This waste include nitrogeneous, gaseous, toxic waste.

Waste includes
- N_2 waste
- Gaseous waste
- Toxic waste (sweat)

Excretory system of human body

2 - Kidney
2 - Ureter
1 - Urinary Bladder
1 - Urethra.



→ Kidney helps in removal of nitrogeneous (N_2) waste by the process of metabolic activities.

anabolism + catabolism $\xrightarrow{\text{simuntaneously}}$ Metabolism.

anabolism	catabolism	Metabolism
simple inorganic substances are combining together to form complex organic substance.	complex organic substance is breaking down into simple inorganic substance.	together anabolism and catabolism is called metabolism

• TRANSPORTATION IN PLANTS

⇒ Raw materials for plants are NPK and other minerals.

Raw materials ← N (nitrogen), P (phosphorus), K (potassium) / other minerals → present in soil, absorbed by roots

→ These are present in soil. They are absorbed by roots.

→ Transportation in plants

•

- leaves prepare food
- roots absorb minerals.

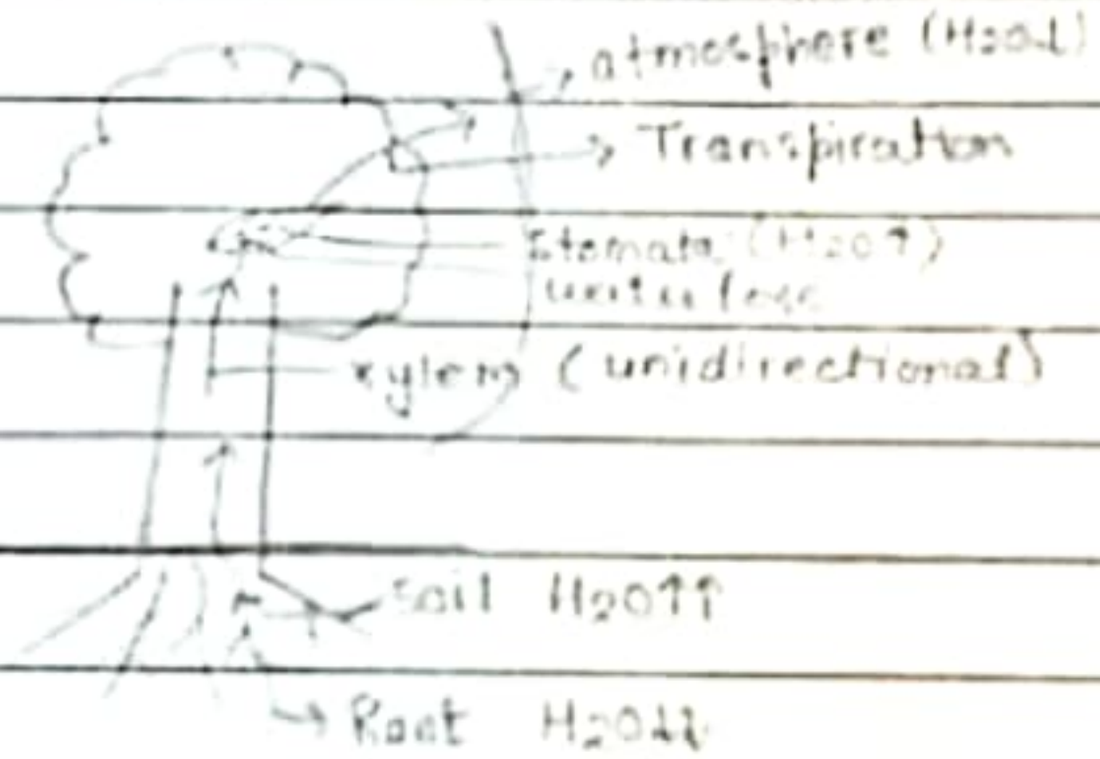
→ Plants do not move so they req. less energy and slow transport system.

→

Conducting duct in plants

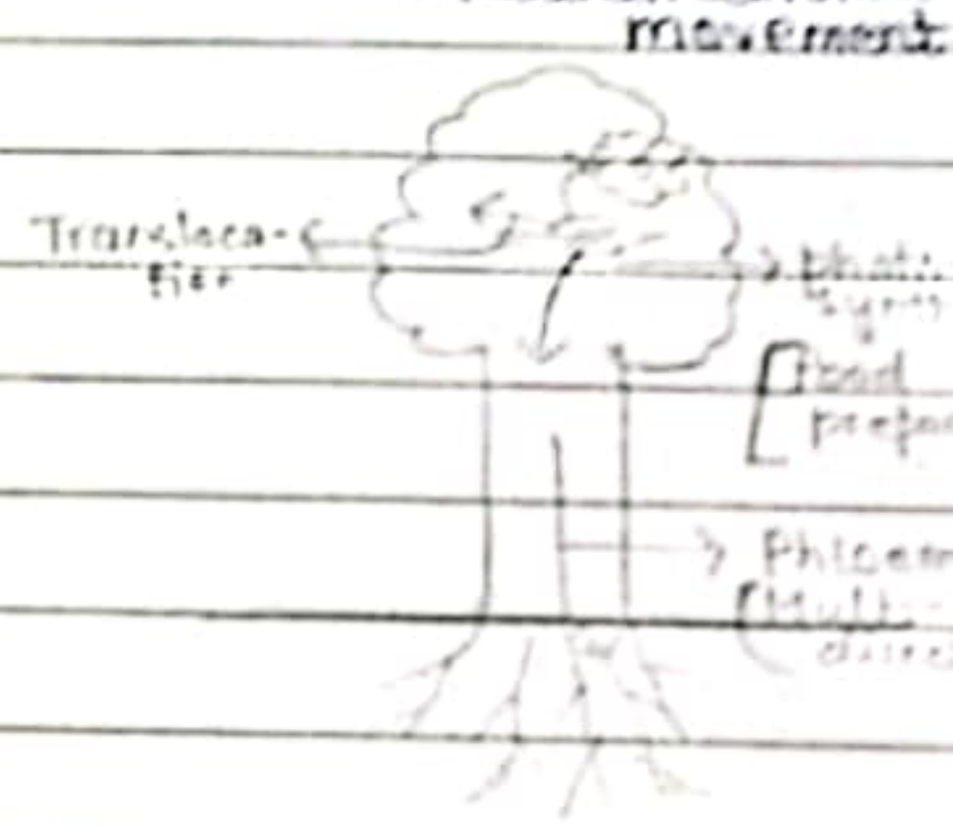
Xylem

- Transports water & minerals.
- Unidirectional movement.



Phloem

- Transport food.
- Multidirectional movement



Xylem
- Vessels
- Tracheids

Phloem
- Food prepared in leaves
- Sieve tubes
- Companion cells

→ In xylem, transportation occurs with the help of simple physical forces.

→ Hence, no energy is required.